

WHY ENGLISH DOESN'T EXTRACT LEFT BRANCHES (YET)

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ABSTRACT

English resists separating pre-head dependents from NP heads: **Whose did you read book?* is sharply unacceptable. Rather than positing a universal constraint, this paper asks what English does instead and measures it. Universal Dependencies rates show available contiguous alternatives, tight determiner–head packaging, and no genuine determiner–head discontinuities in the searched fronting domain. The central claim is distributional: English input gives learners abundant evidence for alternatives that preserve contiguity and no positive evidence for determiner–head discontinuity. The psycholinguistic and functional account is presented as a proposed mechanism, not a demonstrated cause: fused-head parsing may make reanalysis costly, and the workaround family may reduce pressure for a discontinuous construction. Absence of evidence, when the search is thorough, is evidence of systematic non-use rather than proof of impossibility.

Keywords: Left-Branch Condition, Bayesian acquisition, distributional learning, functionalism, corpus linguistics.

I INTRODUCTION

English makes no semantic mystery of *whose* in (1), but the stranded-noun variant (1b) is sharply unacceptable:

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|-----|----|----------------------------------|---------------|
| (1) | a. | <i>You read whose book?</i> | [in-situ] |
| | b. | <i>*Whose did you read book?</i> | [LBE] |
| | c. | <i>Whose book did you read?</i> | [pied-piping] |
| | d. | <i>Whose did you read?</i> | [fused-head] |

As an echo question, the in-situ form (1a) is fine; for ordinary information-seeking questions English pied-pipes the noun (1c). What fails is fronting *whose* while leaving *book* behind (1b). When the noun is omitted, *whose* serves as a fused-head NP (1d), with no overt noun expected.

The point is general: English resists splitting a pre-head dependent from the noun it depends on. Compare:

- | | | | |
|-----|----|--------------------------------------|---------------|
| (2) | a. | <i>She earned how much money?</i> | [in-situ] |
| | b. | <i>*How much did she earn money?</i> | [LBE] |
| | c. | <i>How much money did she earn?</i> | [pied-piping] |
| | d. | <i>How much did she earn?</i> | [fused-head] |
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- | | | | |
|-----|----|--|-----------|
| (3) | a. | <i>I don't think you liked his food.</i> | [in-situ] |
|-----|----|--|-----------|

- b. * *His, I don't think you liked food.* [LBE]
- c. *His food, I don't think you liked.* [topicalized NP]
- d. *His, I don't think you liked.* [fused-head]

These are instances of what is traditionally called **LEFT-BRANCH EXTRACTION** (LBE): the fronting of a pre-head dependent (possessor, quantifier, or degree phrase) while the head remains in situ. Calling it **LEFT-BRANCH EXTRACTION** names the missing behaviour without explaining why English keeps its left branches on a short leash.

Why is English so resistant to LBE? Is the gap a universal structural prohibition, a constraint that holds across languages, or an English-specific pattern emerging from how English packages its nominals?

The standard move in generative syntax is to posit a constraint: the Left-Branch Condition, or some descendant of it (Ross, 1967). But stipulating a ban doesn't explain why the ban exists, or why English obeys it so strictly when other languages appear more permissive. A constraint is a good diagnostic but a poor causal story. For other "island types" (a useful descriptive grouping, if not a natural one), functionalist accounts have made progress: Culicover et al. (2022) tie unacceptability to processing complexity (cf. Gibson, 1998; Hawkins, 2004, on dependency locality); Cuneo and Goldberg (2023) show that island effects correlate with discourse function. But LBE remains underexplained. The Left-Branch Condition doesn't satiate like other islands, and no discourse-functional account of the determiner–head bond has been established.

Instead of asking "what rule blocks LBE?", this paper asks: what does English do instead? The answer is first of all distributional. English input supplies abundant contiguity-preserving alternatives and, in the searched domain, no genuine cases of determiner–head discontinuity. The psycholinguistic and functional pieces are offered as a proposed mechanism for why such a distribution could be learned and remain stable.

The **DISTRIBUTIONAL** strand is straightforward. English makes heavy use of **FUSED-HEAD** nominals (constructions where a determinative or modifier realizes the head function, leaving no overt noun). And English has a family of productive constructions that express degree modification and fronted-determiner dependency formation without separating determiner from head: pied-piping (*whose book*), the *big mess* construction (*how good a teacher*), *of*-insertion (*how big of a risk*), and predeterminer modifiers (*what a mess*). These workarounds are frequent and entrenched (cf. Bybee, 2010, on entrenchment); hypothetical LBE is not. The corpus data shows that workarounds occur at measurable rates while LBE-like configurations are essentially absent.

The proposed **PSYCHOLINGUISTIC** strand follows from the first. When a speaker fronts *whose*, a listener's first parse is likely to treat it as a complete NP (*whose* = 'whose one'), not as a displaced dependent. Fused-head productivity therefore provides a plausible route by which the distributional facts could bias comprehension toward a fused-head construal. On this view, an LBE parse would require reanalysis.

The proposed **FUNCTIONAL** strand is complementary. The workaround family does much of the semantic work that LBE would do. This makes it plausible that no strong communicative pressure favours a new discontinuous construction; the corpus evidence by itself does not prove that causal history.

The strategy is to measure, not judge. The stars in (1)–(3) have already done plenty of judging; the question here is what English does with its time instead. Rather than collecting acceptability rat-

ings for starred sentences, this paper estimates rates: how often do these workaround constructions occur, how tight is determiner–head packaging, and how rare are configurations that would require an LBE parse? The aim is English-internal. The data comes from English usage, and the causal interpretation is deliberately modest: distributional facts are demonstrated; processing and functional explanations are proposed mechanisms. This framing brackets cross-linguistic comparison. Slavic languages are often said to “allow LBE”, but whether the Slavic pattern and the English gap instantiate the same structural configuration is far from clear. LEFT BRANCH is a descriptive class tied to a particular phrase-structure analysis, not a valid comparative concept (Haspelmath, 2010). Following Croft (2001), this paper treats the English pattern as English-specific, without assuming that discontinuous nominals in case-rich, article-less languages are commensurate.

The logic is Bayesian: if learners receive no positive evidence for determiner–head separation, a model of acquisition should assign little support to it. The gap can then be maintained without positing that grammar makes LBE impossible. The familiar aphorism says that absence of evidence is not evidence of absence. But when the opportunity set is large and the search is thorough, absence of evidence is evidence of systematic non-use, not proof of impossibility. That distinction matters because systematic non-use can change.

2 THE CONSTRUCTIONS

Before operationalizing anything, it helps to see the constructions tracked here. Table 1 shows the main patterns, with examples. In the table, “int./rel./excl.” means interrogative, relative, or exclamative.

TYPE	PATTERN	EXAMPLES
Pied-piping	interrog./rel./excl. + N	<i>whose book, which car</i>
Fused-head	interrog./rel./excl. alone (headless)	<i>whose is better, which did you choose</i>
<i>big mess</i>	degree + Adj + <i>a</i> + N	<i>how good a teacher, so big a deal</i>
<i>of</i> -insertion	degree + Adj + <i>of a</i> + N	<i>how big of a risk, too good of a deal</i>
Predeterminer modifier	modifier + <i>a</i> + N	<i>what a mess, such a deal</i>

Table 1: English constructions relevant to LBE.

The label *big mess* is informal but apt: these are precisely the places English tidies up the semantics while refusing to untie determiner from head.

What unites these patterns is that they do the semantic work that hypothetical LBE would do, expressing degree, quantity, or possession in interrogatives and exclamatives while keeping determiner and head contiguous. These are not marginal constructions: Sommerer (2022) shows that *big mess* patterns like *so great a desire* were far more frequent in Early Modern English than they are today, and Van Eynde (2007) provides a formal HPSG account of their synchronic structure.

Just how tight is determiner–head packaging? Jackendoff (2025) argues that full parenthetical clauses can’t interrupt, e.g.:

- (4) * *The, I think, report was leaked.*

A search of our corpora (see Section 4) for determiner–head arcs with intervening punctuation, followed by manual inspection for parenthetical supplements, yields no clear cases of phrase-sized inter-

ruptions.¹ Jackendoff's explanation is prosodic: parentheticals require Intonational Phrase boundaries, and a determiner and its head form a tight prosodic unit (typically a clitic group or single Phonological Phrase) with no internal boundary where a supplement could attach (Jackendoff, 2025, pp. 33, 42); for the prosodic hierarchy that makes this prediction, see Selkirk (1984) and Nespor and Vogel (2007). If the bond resists even parenthetical phrases, extraction across the same bond would be unexpected.

This parsing bias is noted by Culicover et al. (2022, p. 24): a clause-initial *which*, *what*, or *whose* form is processed by default as a phrasal argument with an elided nominal head, that is, as a fused-head NP. The result is a garden-path effect when the listener encounters a bare noun with no determiner. English fused-head productivity, then, does double duty: it provides an alternative to LBE in production, and it biases us against an LBE construal in comprehension.

This bias has a narrow domain. It can operate only where the fronted form has an independent nominal use. The relevant forms include *which*, *what*, and, in suitable contexts, *whose*: each can be parsed as a complete NP before any later head is encountered. The same is not true of all determinatives. *Every* and prenominal *no*, for example, require an overt nominal head in dependent use (*every book*, *no money*); they cannot themselves head an NP in the relevant sense (**I read every*, **I bought no*); and parallel LBE patterns are no better.² For *every*, the independent nominal alternative is supplied by *every one* rather than by *every* itself; for *no*, it is supplied by *none*, with dependent *no* paired with independent *none* much as dependent *my* is paired with independent *mine*.

These cases remove one component of the account. Since *every* and prenominal *no* are not available as fused-head NPs, a fronted *every* or *no* would not initially be parsed as a complete argument in the way a fronted *which*, *what*, or *whose* can be. Yet the determiner–head split is still unavailable. The fused-head parsing bias is therefore not the whole explanation. What survives across the full set is the distributional and functional part of the account: learners encounter no positive evidence for determiner–head discontinuity, and English already supplies contiguity-preserving alternatives, either by pied-piping the nominal or by using a dedicated independent form. *None* is the limiting case: it gives English an independent negative nominal and thereby removes much of the functional pressure that might otherwise favor LBE of dependent *no*.

The dependent/independent morphology points in the same direction. English often distinguishes dependent determiner forms from independent forms: *no* versus *none*, *my* versus *mine*. This split does not by itself prove a syntactic boundary, but it is exactly what one expects in a system where the determiner–head relation is a well-entrenched packaging unit. Morphology and syntax point the same way.

3 A BAYESIAN PERSPECTIVE ON ACQUISITION

The acquisition question here isn't whether a learner can entertain some abstract possibility like "a pre-head dependent separates from its head". It is what analysis the learner assigns to the kinds of strings English actually provides (cf. Tomasello, 2003, on usage-based acquisition). In the domain at issue, English input supplies two analyses that are both frequent and that both keep determiner and

¹The search filtered det-arc candidates with intervening punctuation and manually inspected for appositives, non-restrictive relatives, and parenthetical clauses. Apparent hits were lists or punctuation inside modifier phrases, not genuine det–supplement–head sequences. Hesitation markers like *the*, *uh*, *report* may occur in speech, but these are disfluencies, not grammaticalized supplements; the corpora here are predominantly written.

²I owe this observation to Peter Culicover (p.c.).

head contiguous: headed NPs with pied-piping (*whose book, how much money*) and fused-head NPs where the fronted form itself realizes head function (*whose did you read?, how much did she earn?*). A hypothetical LBE analysis would add a third option: the fronted item is a displaced dependent whose head noun will appear later without it.

What makes absence informative here is that the opportunity set is large. Fronting is everywhere in English input: open interrogatives, relative clauses, exclamatives, topicalization, and focus fronting. A learner repeatedly encounters contexts where an LBE-style analysis would be usable and diagnosable, contexts where a fronted form could, in principle, be parsed as a displaced dependent of a later noun. So the learner doesn't face an undifferentiated "infinite hypothesis space" in which negatives are unlearnable. They face a neighbourhood of analyses in which one extension is a natural next step but is systematically absent from exactly the contexts where it would be licensed.

Given this setup, a Bayesian framing isn't window dressing but a compact way of stating the inference problem (cf. Goldwater et al., 2009; Xu & Tenenbaum, 2007, for Bayesian approaches to word learning and segmentation). The intuition is simple: beliefs should reflect evidence. A learner who has encountered a thousand opportunities for LBE and zero instances has strong evidence that LBE doesn't happen. Not proof (you can't prove a negative), but strong evidence. The more opportunities, the stronger the evidence. This is what Bayesian reasoning formalizes: your confidence in a hypothesis should track how well it predicts what you actually observe (Perfors et al., 2011; Yang & Piantadosi, 2022). If your grammar predicts that LBE should sometimes occur, but it never does, that grammar fits the data poorly. A grammar that predicts determiner-head contiguity fits better.

Two distributional facts push hard in the same direction. First, fused-head nominals are genuinely productive in English. That means that when a learner hears a fronted *which/what/whose* in an interrogative or relative, the parse 'this is already a complete NP' has substantial prior support from experience. Second, the headed alternative that English also supplies is pied-piping, which again preserves determiner-head contiguity. In that setting, an LBE analysis isn't merely unattested; it is a structurally costlier commitment. It asks the learner to treat the same surface cue (fronting) as licensing a dependency whose downstream signature is determiner-head discontinuity with an overt nominal appearing later, exactly the configuration that competes with the highly entrenched fused-head parse. This makes a processing mechanism plausible: fused-head productivity offers a production alternative and may bias comprehension away from an LBE construal.

LBE is *on the table* for an English learner in this sense: it is assigned non-zero prior probability because it is a locally conceivable extension of a richly attested strategy, not an alien operation with no foothold in the grammar (cf. Perfors et al., 2011, on hypothesis-space constraints). As an analogy for what it looks like when the input supplies no nearby scaffolding, consider productive noun incorporation of the Mohawk type, which involves building a lexical noun root into the verb word itself, yielding a single finite predicate with a non-referential incorporated nominal:

- (5) *Kanekwarínyu wa'-k-akya'tawi'tsher-ú:ni*
 it.dotted FACT-1SG-dress-make
 'I dress-made a polka-dotted one.' (Mohawk; Mithun 1984)

English can of course express the propositional content, but it doesn't supply a productive schema in which incorporation systematically alternates with a transitive clause in a way that tracks the specific/non-specific contrast typical of incorporation systems. Lexicalized V-N compounds like *babysit* or *spoon-feed* don't provide the same constructional logic. In Bayesian terms, a learner may be able to

represent such a grammar in principle (incorporation isn't logically impossible), but the prior mass assigned to it is vanishingly small because English input provides no scaffolding constructions from which it could be inferred. LBE is different: the analogues (fronting, pied-piping, fused-head construals) are abundant. What the learner may acquire, then, is not a rule but a norm: we don't do that.

That last clause matters. The proposed acquisition target is not an abstract theorem about "English" but a community norm about packaging. If a learner has spent years hearing determiners and prenominal dependents appear tightly integrated with their heads, and has heard fronting realized overwhelmingly through pied-piping or fused-head NPs, then a string like **Whose did you read book?* violates an entrenched community expectation. The corpus evidence does not by itself prove that this is the source of the felt unacceptability; it shows that the distribution needed for such a norm is present. This kind of account doesn't require a derivational ban. It requires only that learners infer domain restrictions on otherwise productive schemas from distributional evidence.

A community-norm framing also makes room for change without weakening the synchronic distributional result. Learners don't treat all input as coming from one homogeneous source; they notice that different people talk differently. Teenagers talk differently from grandparents; Twitter is different from academic prose. A learner tracking "how often does this structure occur?" is really tracking "how often does this structure occur *among people like X?*" For mainstream English speakers, the answer for LBE is: never, as far as I can tell. But if a subgroup started using LBE (if it became a marker of some community), a learner could notice that and track it separately. The account is compatible with language change: what shifts is not a universal constraint but which community's patterns the learner is tracking. This is also why pooling across corpora makes sense in the analysis below: the learner is effectively averaging over sources, weighting each by how much data it provides.

This perspective also suggests a way to interpret a puzzling experimental finding. Syntactic SATIATION is the tendency for repeated exposure to make initially unacceptable sentences more acceptable (Snyder, 2000). Many island violations show satiation: hear them enough times in a lab, and they start to sound better (Lu et al., 2024). But LBC violations don't satiate. Lu et al. (2024), in a meta-analysis of 25 experiments, found no satiation effect for left-branch extraction. The distributional account predicts such resistance to brief exposure: if a learner has heard tens of thousands of determiner-head sequences and zero LBE cases, a handful of **Whose did you read book?* tokens in an experiment should not move expectations much. This is a compatibility claim, not direct processing evidence.

This framing makes the empirical question explicit. If the Bayesian account is right, we should find:

1. Workaround constructions at non-trivial rates relative to their opportunities.
2. LBE-like configurations at near-zero rates relative to their opportunities.
3. Tight determiner-head packaging, with little or no intervening material.

The next section describes how to measure these rates.

4 DATA AND METHOD

The predictions above are empirical, and testing them requires measuring rates, not raw counts but relative frequencies. Saying "the *big mess* construction is frequent" means nothing without a denominator: frequent relative to what?

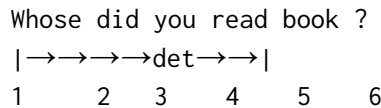
Corpus data can test this directly. Unlike acceptability judgments, which ask whether a structure *could* occur, corpus rates tell us what *does* occur, and so what learners are exposed to. The goal is to estimate, with uncertainty, the rate of each construction given its opportunity set.

The opportunity set differs by construction. For the *big mess* construction (*that big a mess, how good a teacher*), the denominator is: how many configurations involve degree modification of an adjective with an indefinite NP? The rate is *big mess* tokens divided by that opportunity count. For fused-head construals (*whose* as a complete NP), the denominator is: how many fronted *which/what/whose* forms occur with an overt head versus without one? For LBE-like configurations: how many determiner–head arcs are there, and how many are discontinuous?

LBE itself can’t be searched in corpora; we can’t find tokens of **Whose did you read book?* because speakers don’t produce them. Instead, the analysis operationalizes what LBE *would* look like if it existed: a determiner separated from its head by intervening material (a verb, a crossing dependency). If such configurations are near-zero, the corpus provides no evidence from which a learner could acquire determiner–head discontinuity.

WHAT AN LBE TOKEN WOULD LOOK LIKE

If English permitted LBE, a sentence like **Whose did you read book?* would require a dependency annotation where *whose* bears a DET relation to *book*, but the arc crosses the finite verb *read*. Schematically:



The DET arc from position 1 to position 5 spans four tokens and crosses a verb. This is the signature: a determiner–head arc in the fronting domain (interrogative, relative, or exclamative fronting) that either (i) spans an intervening verb, or (ii) crosses another dependency arc. If English speakers productively used LBE, such configurations would appear at non-trivial rates in the fronting subset. They don’t. The search below operationalizes exactly this: fronted determiners whose head noun is separated by a verb or by a crossing arc.

The data comes from Universal Dependencies (UD; Nivre et al., 2020), a cross-linguistic annotation standard that provides consistent syntactic structure across corpora. The UD slice comprises sixteen English corpus directories, fourteen of which contribute determiner–head arcs in the scan, spanning edited and unedited registers: newspaper, web text, blog posts, dialogue, learner English, and more. UD is used diagnostically, not as a theory of English structure; every pattern is checked against standard terminology. The search procedure is high-precision by design: it is tuned to produce clean denominators rather than exhaustive recall.

Workaround operationalizations are defined in dependency and order terms. The *big mess* construction is operationalized as degree modifier + adjective + *a/an* + noun in that order, as in *how good a student*. The *of*-insertion variant is parallel but includes *of*, as in *how big of a risk*. The predeterminer modifier cluster captures *such a*, *what a*, and *many a*.

To approximate fused-head availability, the analysis uses UD DET vs PRON tags for *which*, *what*, and *whose*. When UD tags one of these words as PRON, it typically reflects a headless (fused-head) construal; when it tags the same word as DET, the word is in determiner function with an overt head. This is a proxy for headless nominal construals, not a claim about lexical category.

Determiner–head packaging is measured by distance (in tokens), with a threshold of $k = 4$: arcs longer than this are flagged as potential discontinuities. The analysis also tracks whether a verb or crossing arc intervenes, as in *the students who arrived yesterday* (verb-intervening) or *the book, which I recommended, on syntax* (crossing). Arcs with verb or crossing interveners are flagged as LBE-like candidates. The threshold was validated (§7) and treated as a sensitivity parameter.

5 WHAT WE'RE MEASURING AND HOW

The goal is to estimate rates, not to pronounce sentences grammatical or ungrammatical, but to ask: how often does each construction occur, given its opportunities? Four rates matter:

1. How often do workarounds (*big mess*, *of*-insertion) occur when the opportunity arises?
2. How often do speakers use fused-head construals (*whose* alone) versus headed ones (*whose book*)?
3. How tight is determiner–head packaging? How far apart can they get?
4. How often do we see the kind of discontinuity that LBE would produce?

Estimating a rate from a single corpus is straightforward: count the hits, divide by opportunities. But corpora differ in size, and small corpora give noisy estimates. If one corpus has 2 hits out of 10 opportunities (20%) and another has 200 out of 1,000 (also 20%), we should trust the second estimate more.

EMPIRICAL-BAYES PARTIAL POOLING handles this (Gelman & Hill, 2007; McElreath, 2020). The idea: let the corpora inform each other. If most corpora cluster around 5% but one tiny corpus shows 50%, that extreme probably reflects noise, not a real difference. Partial pooling pulls extreme small-sample estimates toward the group average while letting large corpora speak for themselves. The result is CREDIBLE INTERVALS: posterior uncertainty intervals for the rate. A 95% credible interval means that, conditional on the model, prior, and observed data, 95% of the posterior mass for the rate lies in that range.

For rare phenomena (like LBE-like configurations), the question shifts from “what’s the rate?” to “how high could the rate possibly be?” If you search 1,000 opportunities and find nothing, the true rate is almost certainly below 0.3%. The rule of thumb: with zero hits, the 95% upper bound is roughly 3 divided by the number of opportunities. This is the sense in which absence of evidence becomes evidence of absence: not proof of impossibility, but a ceiling on how common the phenomenon could be.

For readers who want to see what the model actually looks like, here it is:

$$\begin{aligned} k_j &\sim \text{Binomial}(N_j, \theta_j) \\ \theta_j &\sim \text{Beta}(\hat{\alpha}, \hat{\beta}) \end{aligned}$$

That’s it. Each corpus j produces k_j hits out of N_j opportunities, with its own rate θ_j . The rates are drawn from a shared Beta distribution with parameters $\hat{\alpha}$ and $\hat{\beta}$. These hyperparameters are estimated from the corpus-level counts and then treated as fixed when forming corpus-level posteriors. No hyperpriors are specified, so the model is empirical Bayes rather than full hierarchical Bayes. It is used as a descriptive rate-estimation device rather than as a model of learners’ mental computation. For the main det–head and fused-head estimates, the results are insensitive to reasonable prior choices; very sparse workaround counts are reported descriptively rather than treated as stable corpus-level estimates (Gelman et al., 2013, ch. 5).

Corpus	<i>big mess</i> k/n (%)	<i>of</i> -insertion k/n (%)	predet- <i>a</i> k/n (%)
atis	0/2 (0.0 [0.0, 66.7])	0/2 (0.0 [0.0, 66.7])	1/1089 (0.1 [0.0, 0.4])
childes	–	–	16/5042 (0.3 [0.2, 0.5])
ctetex	–	–	0/191 (0.0 [0.0, 1.3])
esl	–	–	–
eslspek	0/1 (0.0 [0.0, 85.3])	0/1 (0.0 [0.0, 85.3])	5/368 (1.4 [0.5, 3.0])
ewt	4/25 (16.0 [5.7, 33.7])	1/25 (4.0 [0.4, 17.2])	52/5342 (1.0 [0.7, 1.3])
gentle	–	–	5/309 (1.6 [0.6, 3.5])
gum	3/28 (10.7 [3.1, 25.9])	2/28 (7.1 [1.5, 21.0])	34/4795 (0.7 [0.5, 1.0])
gumreddit	–	–	–
lines	8/18 (44.4 [23.7, 66.8])	0/18 (0.0 [0.0, 12.9])	21/2919 (0.7 [0.5, 1.1])
littleprince	–	–	2/129 (1.6 [0.3, 4.9])
partut	0/9 (0.0 [0.0, 23.8])	0/9 (0.0 [0.0, 23.8])	5/952 (0.5 [0.2, 1.1])
pcedt	–	–	–
pronouns	–	–	–
pud	0/2 (0.0 [0.0, 66.7])	0/2 (0.0 [0.0, 66.7])	3/444 (0.7 [0.2, 1.8])
unidive	–	–	–
Total	15/85 (17.6 [10.7, 26.8])	4/85 (4.7 [1.6, 10.8])	144/21580 (0.7 [0.6, 0.8])

Table 2: Observed workaround counts and raw percentages, with Jeffreys 95% intervals as uncertainty guardrails. Rates are percent of opportunities; the sparse *of*-insertion rows are descriptive rather than stable corpus-level estimates.

The payoff of partial pooling is modest but useful: it lets small corpora contribute without letting their noise dominate, while raw counts remain visible for sparse cases.

6 RESULTS

Across the scanned UD slice, the search finds 86,098 determiner–head arcs and 1,228 fronting-domain opportunities where LBE would be diagnosable. It finds no genuine LBE-like cases after manual inspection. The workarounds are attested; the determiner–head gap is real.

Across the UD corpora, there are 15 *big mess* tokens and 4 *of*-insertion tokens against 85 degree+adjective+*a/an*+noun opportunities. The predeterminer modifier cluster yields 144 tokens: *such a, what a, many a*. The denominator is 21,580 *a/an* determiner–head arcs. The fronted-form proxy yields 1,228 fronted determiner arcs and 6,429 PRON-tagged tokens overall, so the fused-head option is well attested in usage even in a conservative UD proxy.

Figure 1 and Table 2 summarize the workaround family by construction using observed counts and raw percentages first, with Jeffreys intervals as uncertainty guardrails. This matters especially for *of*-insertion, where only four tokens are present. Figure 2 unpacks the same counts by corpus, showing register variation without pretending that sparse cells support precise corpus-level rates.

Determiner–head packaging is tight. Table 3 summarizes the raw counts for LBE-like diagnostics. The critical row is the last: after manual inspection, none of the 1,228 fronted determiner arcs is a genuine LBE-like case. The two verb/crossing hits are diagnostic flags, not LBE tokens. Under the partial-pooling model, the 95% upper bounds on the fronted determiner verb/cross flag rate are below 0.5% in every corpus, typically around 0.3–0.4%.

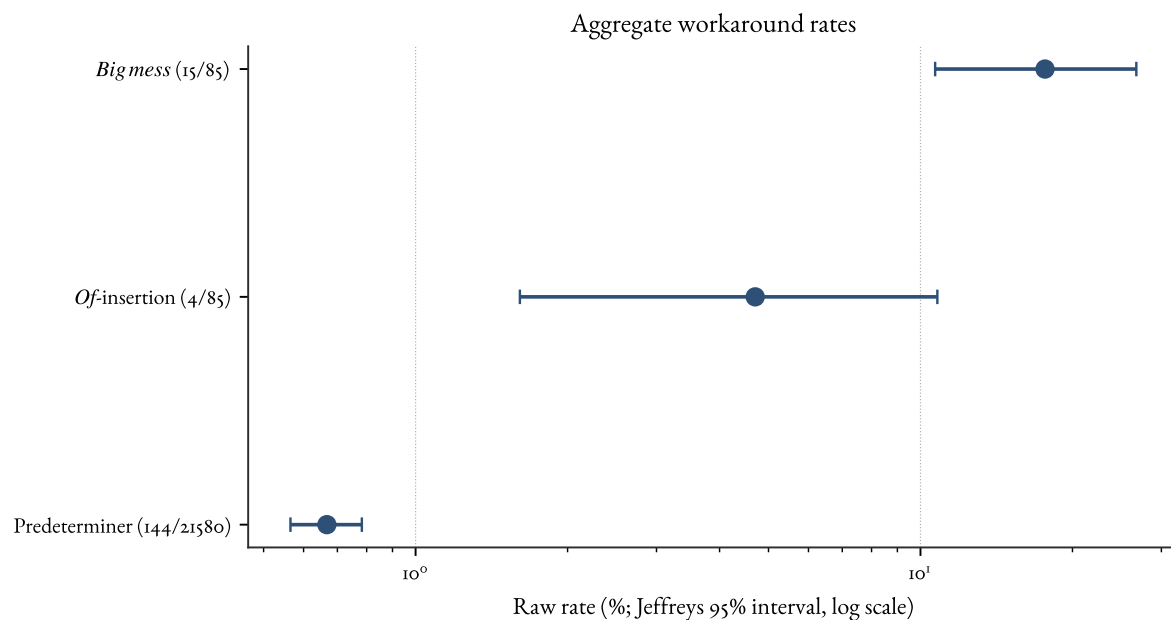


Figure 1: Observed aggregate workaround rates. Points show raw percentages; lines show Jeffreys 95% intervals as uncertainty guardrails. Labels report hit/opportunity counts.

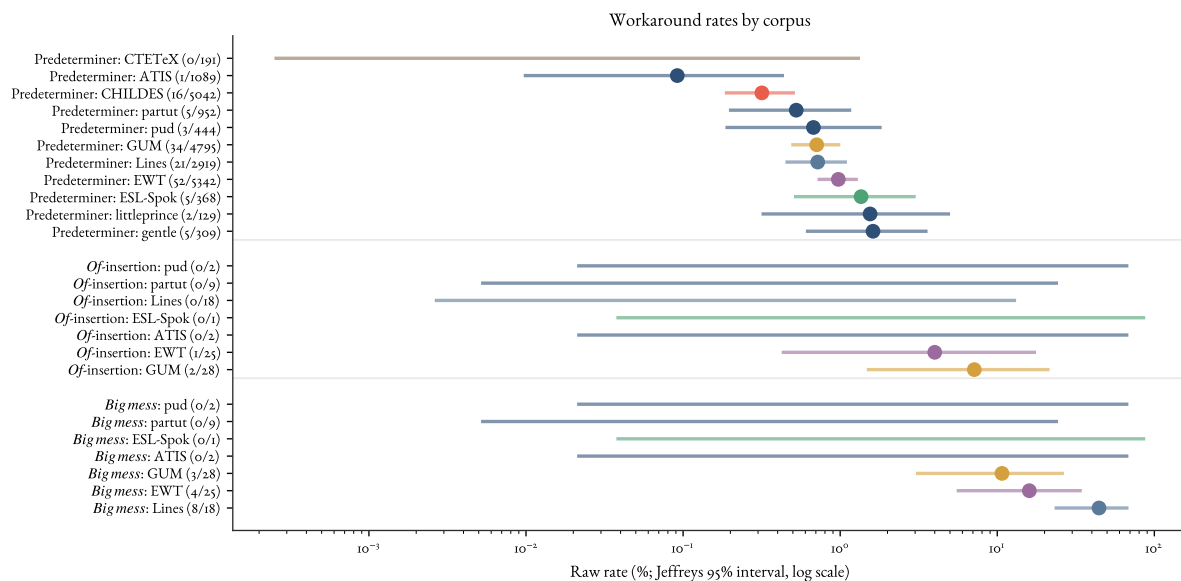


Figure 2: Workaround rates by corpus. Points show raw percentages; lines show Jeffreys 95% intervals as uncertainty guardrails.

Configuration	k (tokens)	N (opportunities)	Rate
All det–head arcs exceeding $k=4$	1,229	86,098	1.4%
...with verb or crossing	279	86,098	0.3%
fronted det arcs exceeding $k=4$	1	1,228	0.1%
fronted det arcs with verb/crossing flag	2	1,228	0.2%
genuine LBE-like cases after manual inspection	0	1,228	0.0%

Table 3: Raw counts for determiner–head discontinuity. The distance and verb/crossing diagnostics are reported separately for fronted arcs; the final row is the manually vetted count of genuine LBE-like cases.

The two fronted determiner hits are *What a moving van!* (CHILDES) and *what a mindblowing servicing* (EWT). Both are exclamatives where a participial adjective (*moving*, *mindblowing*) triggers the verb flag. They are diagnostic false positives rather than genuine LBE-like discontinuities. Genuine LBE-like cases after manual inspection: 0/1,228.

Figure 3 makes the candidate funnel visually explicit: from 86,098 determiner–head arcs, we narrow to 1,228 fronted det arcs, and then to 2 verb/crossing flags, both of which are false positives. The separate distance-only diagnostic contributes 1 fronted det arc above $k=4$.

A natural objection is that NPs are simply islands, full stop. But English does tolerate extraction from NP-internal positions, specifically from PP complements of nouns. The corpora contain examples like *Who’s that a picture of?* and *What is that a picture of?*, where the fronted form has been extracted from the *of*-complement of *picture*, stranding the preposition inside the NP. These aren’t marginal or archaic; they are ordinary conversational questions. The contrast is instructive: PP-complement extraction exists at non-trivial rates, while determiner–head discontinuity stays at zero. If NPs are “islands”, English permits PP-complement extraction while still preserving the determiner–head packaging unit.

Table 4 and Table 5 show the posterior rates for fused-head choices, det–head tails, and verb or crossing diagnostic flags. The flag-rate estimates are not manually vetted LBE rates; the manually inspected genuine count is 0/1,228. Figure 4 complements those rates with det–head distance distributions for headed arcs only, and Figure 5 shows that workaround constructions also preserve tight det–head packaging.

The headed and headless rates are complementary by construction: a fronted-form token is coded as either headed or headless in the local proxy. This complementarity is what we expect if headed and headless fronted forms occupy the same local choice space. Whether speakers experience them as competing options remains an interpretation of the distribution, not something the correlation independently proves.

These numbers don’t imply impossibility. They imply that LBE-like configurations, as currently operationalized, are exceedingly rare relative to their opportunity sets, while alternative packaging is available and frequent. In-situ fronted determiners (echo questions, embedded interrogatives) are sparse in these written-dominant corpora, but the handful that appear show the same tight det–head packaging, consistent with the main finding, though too few to estimate rates reliably.

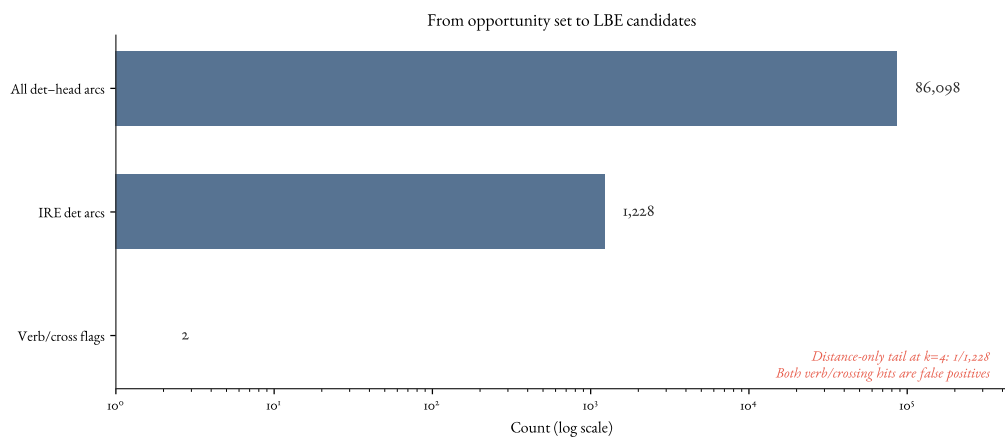


Figure 3: From opportunity set to LBE candidates. The funnel narrows from 86,098 det-head arcs to 2 verb/crossing flags, both false positives (participial adjectives).

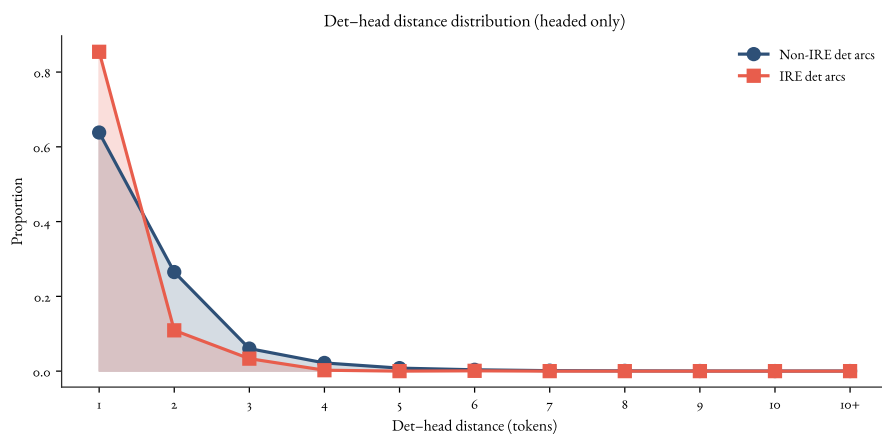


Figure 4: Det-head distance distribution for non-fronted determiner arcs versus fronted determiner arcs (headed only).

Corpus	fused-head (%)	fronted det verb/cross flag (%)
atis	57.209 [54.784, 59.635]	0.131 [0.030, 0.306]
childes	88.475 [87.328, 89.589]	0.202 [0.060, 0.427]
ctetex	70.213 [47.569, 88.379]	0.179 [0.041, 0.416]
esl	–	–
eslspek	78.353 [67.750, 87.482]	0.178 [0.040, 0.412]
ewt	89.470 [87.543, 91.264]	0.218 [0.064, 0.462]
gentle	95.890 [90.533, 99.022]	0.178 [0.043, 0.414]
gum	93.897 [92.404, 95.239]	0.173 [0.041, 0.400]
gumreddit	–	–
lines	92.200 [89.632, 94.421]	0.176 [0.041, 0.404]
littleprince	89.662 [80.043, 96.432]	0.178 [0.040, 0.411]
partut	96.529 [93.814, 98.461]	0.179 [0.041, 0.418]
pronouns	–	–
pud	95.643 [90.785, 98.756]	0.179 [0.040, 0.414]

Table 4: Empirical-Bayes posterior mean and 95% interval for fused-head rates and fronted determiner verb/cross flag rates. Rates are percent of opportunities; the flag rate is diagnostic, not a manually vetted LBE rate.

7 MEASUREMENT ERROR AND SENSITIVITY

Any corpus search can misfire: it may flag tokens that don’t actually instantiate the target construction (false positives), or miss tokens that do (false negatives). To quantify this for the rate estimates, a stratified random sample of 286 rows was drawn from the $k=4$ output (covering flagged hits, near-misses, and clear non-hits), and each row was manually annotated for whether it genuinely instantiated the target pattern. This yields two error rates for the validation sample: the FALSE DISCOVERY RATE (FDR), the proportion of flagged hits that are wrong, and the FALSE OMISSION RATE (FOR), the proportion of non-hits that should have been flagged. The observed validation-sample FDR is 0.0%; the observed FOR is 2.4%. With a $\text{Beta}(1,1)$ prior these correspond to posterior means of 0.6% and 3.2%. The false negatives are concentrated in exclamatives (*what a N*) plus a single *big mess* token, which makes a uniform correction conservative for aggregate rates. The two fronted verb/crossing diagnostic flags discussed above were manually adjudicated separately and are excluded from the genuine LBE count.

The error model is used as a sensitivity analysis rather than a primary estimate. The posterior on the induced true-hit counts is reported per corpus, which lets readers map back from rates to concrete counts. These adjustments affect workaround and diagnostic-flag rates; the central LBE result rests on the manually inspected count of 0 genuine cases out of 1,228 fronting opportunities.

8 ROBUSTNESS CHECKS

The det-head tail is stable across threshold choices. Re-running with $k = 2$ and $k = 3$ leaves the fronted determiner verb/cross flag upper bounds below 0.5% in every corpus (max 0.42%). A corpus hold-out posterior predictive check flags a small set of corpus-specific extremes (including ATIS fused-head rates, LINES *big mess*, and CTETEX/CHILDES det-arc tails), which points to register-specific

Corpus	det-arc tail (%)	det-arc verb/cross flag (%)
atis	0.781 [0.557, 1.047]	0.037 [0.004, 0.105]
chldes	0.088 [0.046, 0.144]	0.021 [0.004, 0.050]
ctetex	5.819 [4.455, 7.360]	1.813 [1.136, 2.642]
esl	0.694 [0.520, 0.893]	0.153 [0.079, 0.250]
eslspek	0.314 [0.090, 0.678]	0.197 [0.040, 0.479]
ewt	2.294 [2.093, 2.507]	0.569 [0.468, 0.679]
gentle	1.858 [1.191, 2.677]	0.208 [0.041, 0.506]
gum	2.095 [1.893, 2.303]	0.473 [0.379, 0.575]
gumreddit	1.661 [1.024, 2.444]	0.624 [0.285, 1.100]
lines	0.795 [0.632, 0.978]	0.118 [0.061, 0.193]
littleprince	0.506 [0.108, 1.178]	0.379 [0.075, 0.919]
partut	1.673 [1.330, 2.055]	0.326 [0.186, 0.504]
pronouns	0.417 [0.014, 1.461]	0.214 [0.003, 0.826]
pud	1.463 [0.997, 2.008]	0.257 [0.094, 0.502]

Table 5: Posterior mean and 95% credible interval for det-head tail rates. Rates are percent of opportunities.

Rate	Corpus	k/n	Tail area
interrog./rel. headless (int/rel)	atis	898 / 1572	0.000
<i>big mess</i>	lines	8 / 18	0.000
fused-head	atis	898 / 1574	0.000
det-arc verb/cross flag	ctetex	20 / 920	0.001
det-arc tail	ctetex	57 / 920	0.002
det-arc tail	chldes	11 / 13739	0.024
predet- <i>a</i>	atis	1 / 1089	0.039
predet- <i>a</i>	chldes	16 / 5042	0.061

Table 6: Lowest posterior predictive tail areas from corpus hold-out checks (diagnostic only).

variation rather than a global misfit. Threshold-specific validation favours $k = 4$: it preserves zero false discoveries while lowering the false omission rate from 14.5% ($k = 2$) to 2.4% ($k = 4$).

Figure 6 summarizes the $k = 2/3/4$ robustness checks, and Table 6 lists the lowest posterior predictive tail areas from the corpus hold-out check (diagnostic only, not a threshold test). Prior-sensitivity checks compare uniform Beta(1,1), Jeffreys Beta(0.5,0.5), and skeptical Beta(1,100) specifications. They yield nearly identical posteriors for the near-zero fronted det verb/cross flag rate; for workaround rates with more data, the posteriors converge regardless of prior.

A final check addresses the obvious question: can the data distinguish “LBE truly impossible” from “LBE rare but possible”? With zero genuine cases out of 1,228 opportunities, the 95% upper bound on the true rate is 0.24%, about 1 in 400. If LBE occurred even that rarely, a sample this size would have a 95% chance of catching at least one case. It caught none. The sample size is sufficient to detect LBE if it occurred at modestly low rates; zero hits isn’t compatible with “rare but possible” unless “rare” means below 0.1%. Figure 7 visualizes this: panel A shows where the true rate probably

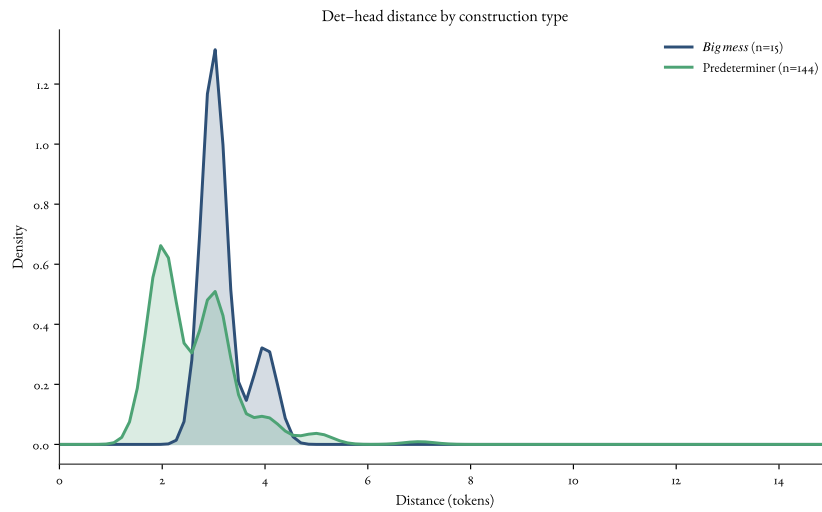


Figure 5: Det-head distance by construction type. Workaround constructions also show tight packing.

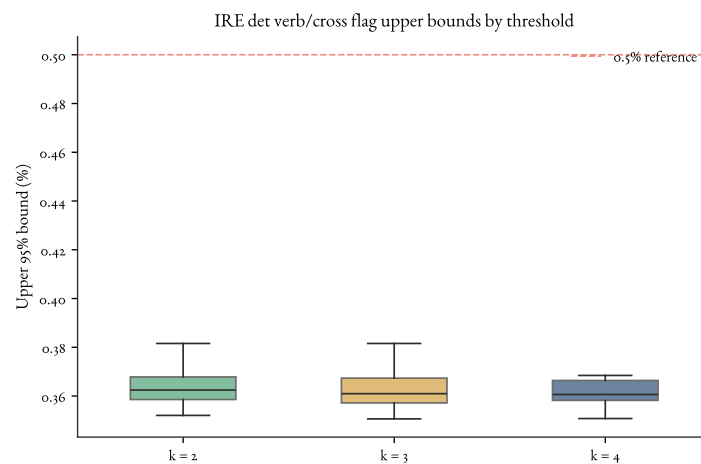


Figure 6: Upper bounds on fronted determiner verb/cross flag rates by distance threshold.

lies given zero hits; panel B shows how unlikely zero hits would be if LBE were actually possible; panel C shows the detection power, how likely we'd be to catch LBE if it existed at various rates.

9 LIMITATIONS AND FUTURE WORK

The Bayesian story rests on two empirical claims: that learners encounter the relevant evidence, and that the workaround family captures the relevant competitor set. Neither is fully demonstrated here.

On the evidence side, the corpora are predominantly written. The determiner-head packaging claim (that even parentheticals can't interrupt) is strongest if it holds in speech, not just in edited punctuation conventions. Spoken corpora could test this with time-aligned transcripts or prosodic annotation, measuring pause and disfluency incidence between determiner and head versus other NP-internal positions. If determiner-head turns out to be especially boundary-resistant even in disfluent speech, the prosodic-tightness argument is strengthened; if not, the claim needs to be reframed as a production preference rather than a structural impossibility.

On the competitor set, the current inventory may be incomplete. Partitives (*which of these books, how many of the students*) occupy a similar communicative niche to pied-piped fronted NPs but route the dependency through a PP. Kind/sort/type-of constructions (*what kind of book*) are another contiguity-preserving alternative. Clefts and pseudo-clefts (*Whose book was it that you read?*) externalize the packaging problem entirely. If these competitors are frequent, the denominator story changes: the absence of LBE is supported not just by pied-piping and fused-head, but by a broader family of schemas that all preserve determiner-head contiguity.

The fused-head proxy (UD PRON vs DET tags) is convenient but imperfect. A gold-annotated sample of fronted strings, coded for fused-head versus headed construal and for local ambiguity, would validate the parsing-bias claim directly.

Finally, the denominators here are global: all determiner-head arcs, all fronting tokens. A more precise Bayesian account would condition on the constructional niche, the contexts where an LBE analysis would be a plausible competitor, not just any determiner. Refining the opportunity set to fronting environments specifically would make the learner's competition set explicit.

A different limit is cross-linguistic, and this paper has imposed it from the outset. Languages often said to permit LBE, including some Slavic languages, are not thereby shown to instantiate the English configuration: LEFT BRANCH is a descriptive class, not a comparative concept. The account still makes a prediction about superficially comparable discontinuous-nominal patterns. Such patterns should be learnable where three conditions converge: recoverable inflectional marking, an independently productive source of discontinuous order, and a discourse niche for the separated element. In languages with the relevant Slavic-style ingredients, case, gender, and number marking could help identify a fronted modifier as belonging with an as-yet-unheard nominal head; productive scrambling or related word-order freedom could supply the constructional scaffolding; and focus-fronting could supply the discourse niche. English has focus-fronting, but it lacks the relevant inflectional recoverability and lacks a productive nominal-discontinuity schema. The same learning logic that leaves the English gap unfilled thus predicts where superficially similar patterns should be learnable, without assuming that the English and Slavic patterns instantiate a single comparative object. Testing that prediction is future work.

A promising next step would connect this distributional story to the discourse-function tradition (Cuneo & Goldberg, 2023). Within the fronting/degree opportunity set, what determines whether

speakers pied-pipe a headed NP, go headless, or choose a workaround? If head omission rises with discourse recoverability and pied-piping rises with head novelty (while det–head discontinuity remains absent regardless), that would link the distributional pattern to discourse pressure. The testable hypothesis is that even when the discourse conditions that Cuneo and Goldberg identify as favourable to extraction are met, English should still route around det–head separation.

10 CONCLUSION

Why don't English speakers say **Whose did you read book?* The data here do not prove the cause of that unacceptability. They do show that the input gives learners no positive evidence for the pattern and abundant evidence for contiguity-preserving alternatives.

The account has three parts, with different evidential status: one demonstrated distributional result and two proposed mechanisms.

DISTRIBUTIONAL: LBE-like configurations are absent from the data. Across 1,228 fronting opportunities, not a single genuine case of determiner–head discontinuity appears. Meanwhile, workarounds (pied-piping, *big mess*, *of*-insertion, predeterminer modifiers) occur at non-trivial rates. A learner encounters abundant evidence for contiguous packaging and zero evidence for discontinuity.

PSYCHOLINGUISTIC: fused-head parsing may commit listeners early. When English speakers hear a fronted *whose*, *which*, or *what*, their default parse can treat it as a complete NP. That commitment would create a garden-path effect if a bare noun appeared later: the listener expected the dependency to be closed, but the speaker is asking for reanalysis. This is a plausible processing mechanism for the distributional pattern, not a result directly tested here.

FUNCTIONAL: contiguity-preserving constructions have established success conditions. Pied-piping, fused-head construals, and the workaround family all meet communicative needs without separating determiner from head. This makes it plausible that LBE offers little independent communicative payoff, but the corpus data do not by themselves prove that functional redundancy caused the gap.

The analysis measures what English speakers actually do, and finds that the alternatives are entrenched while LBE-like configurations are essentially absent. A learner acquiring English has zero observed evidence for discontinuity and abundant evidence for packaging in the searched domain.

What the analysis documents is the synchronic state: a distribution in which the relevant tokens are absent and the alternatives are present. This can support a self-perpetuating learning loop: no tokens, so no learning; no learning, so no production; no production, so no tokens. That loop would explain stability but not origin. A fuller causal account would need processing, production, or discourse evidence showing that LBE is costly in a way that its competitors are not. Jackendoff's IP-Break Hypothesis identifies a possible structural path for this story: a prosodic bottleneck through which such pressure could manifest. If that functional pressure operated in an earlier stage of English, it could have produced the statistical situation now observed. Once established, the distribution could maintain itself even without ongoing functional pressure.

But the distribution is *contingent*, not necessary. A constraint-based account would say LBE *cannot* emerge in English; there is a grammatical principle blocking it. The functionalist proposal says only that it *has not*; the social and statistical conditions have not favored it. This is a distinguishing prediction.

The contingency claim still matters. If there is no independent grammatical ban, a discourse com-

munity could conventionalize a discontinuous pattern, but only if learners receive stable, identifiable evidence for it and some constructional niche makes the pattern useful. The present data shows no such evidence in mainstream written English. The prediction is modest: convention can change, but without community-specific input and a constructional niche, learners have no reason to posit determiner–head discontinuity.

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<https://github.com/BrettRey/English-LBC-functionalist>.

A CORPUS INVENTORY

Table 7 lists the UD corpus directories scanned, whether they contributed determiner–head arcs, and which analyses they entered. The two directories with zero determiner–head arcs were still scanned, but they do not contribute to rate estimates.

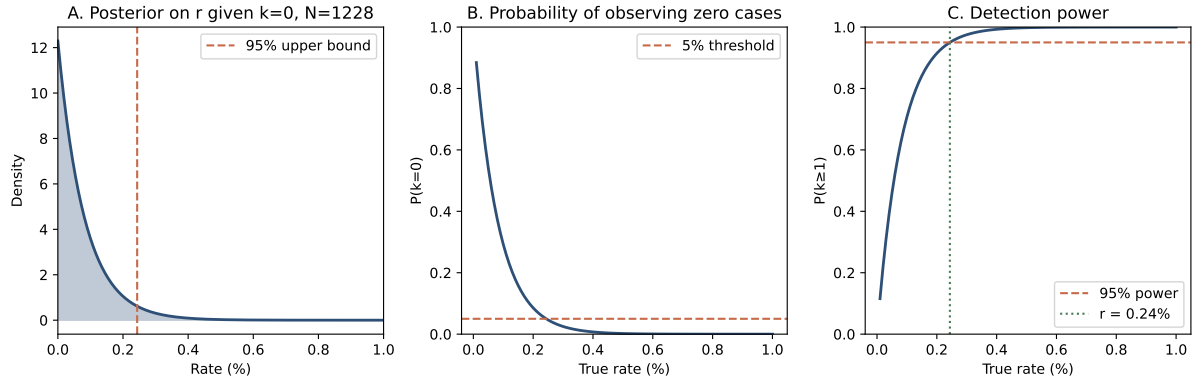


Figure 7: Distinguishing “impossible” from “rare”. (A) Where the true rate probably lies, given zero hits. (B) How likely zero hits would be at various true rates. (C) Detection power: how likely we’d catch at least one case.

Corpus	Det arcs	Fronted det arcs	Analyses entered
atis	4,809	692	det-tail; fronted/fused; workaround
childes	13,739	292	det-tail; fronted/fused; workaround
ctetex	920	4	det-tail; fronted/fused
esl	7,569	0	det-tail
eslspek	1,233	13	det-tail; fronted/fused; workaround
ewt	19,597	112	det-tail; fronted/fused; workaround
gentle	1,167	2	det-tail; fronted/fused; workaround
gum	18,563	66	det-tail; fronted/fused; workaround
gumreddit	1,197	0	det-tail
lines	9,868	36	det-tail; fronted/fused; workaround
littleprince	540	4	det-tail; fronted/fused; workaround
partut	4,651	5	det-tail; fronted/fused; workaround
pcedt	0	0	scanned only
pronouns	190	0	det-tail
pud	2,055	2	det-tail; fronted/fused; workaround
unidive	0	0	scanned only

Table 7: Corpus inventory for the UD scan. “Det-tail” means the corpus contributes to determiner–head distance and verb/crossing diagnostics; “fronted/fused” means it contributes to the fronted-form proxy; “workaround” means it contributes at least one relevant workaround opportunity.

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